DS-A3: Regression Baseline and Diagnostics

Aim Build a transparent regression baseline for daily solar-wind speed using a frozen, time-ordered split and inspect where prediction errors may be unreliable.

Data and method The supplied OMNI daily text file has 23,376 rows and 55 columns. The target is solar-wind speed (km/s); predictors are proton density, IMF magnitude, GSM Bz, and flow pressure. Sentinel/fill values are treated as missing. Rows with missing target or selected predictors are removed, dates are ordered, and duplicate dates are dropped. The resulting analysis has 19,019 usable rows: 13,313 training, 2,853 validation, and 2,853 chronological test rows.

The naive baseline predicts the training median. Candidate models are ordinary linear regression, ridge (alpha=1), and lasso (alpha=0.01), using median imputation and standardization. The model is selected by validation RMSE; the test split is reserved for final evaluation.

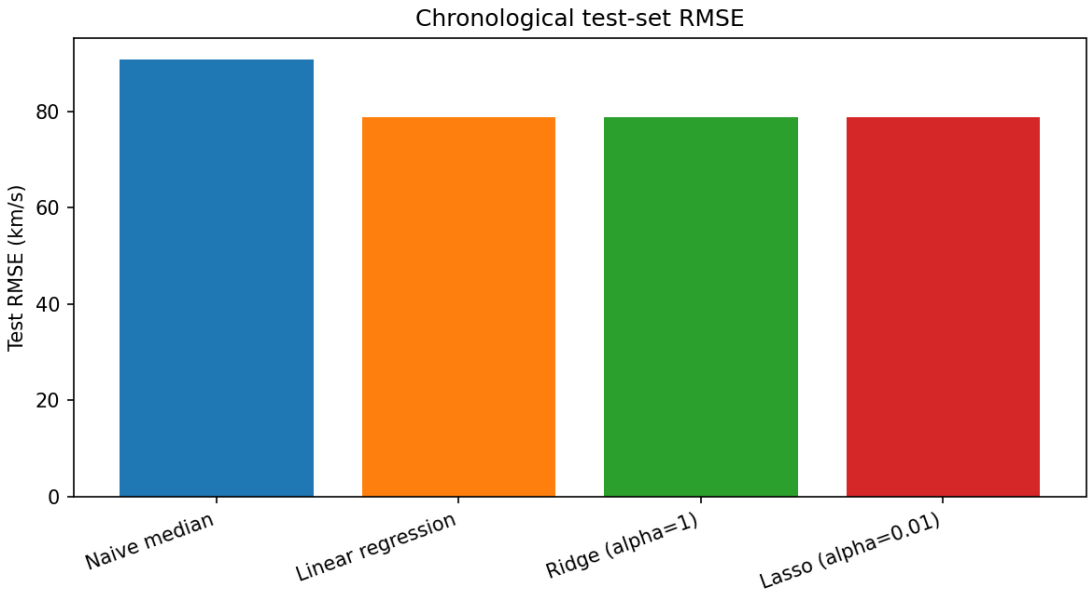
Results The validation-selected model was **Linear regression. On the held-out chronological test set, it achieved MAE=65.453 km/s, RMSE=78.812 km/s, and R^2=0.244 (n=2,853). The naive median baseline on the same test set achieved MAE=70.736 km/s, RMSE=90.670 km/s, and R^2=-0.001. Full comparison metrics are in `model_comparison.csv`; test predictions and residual slices are also supplied.**

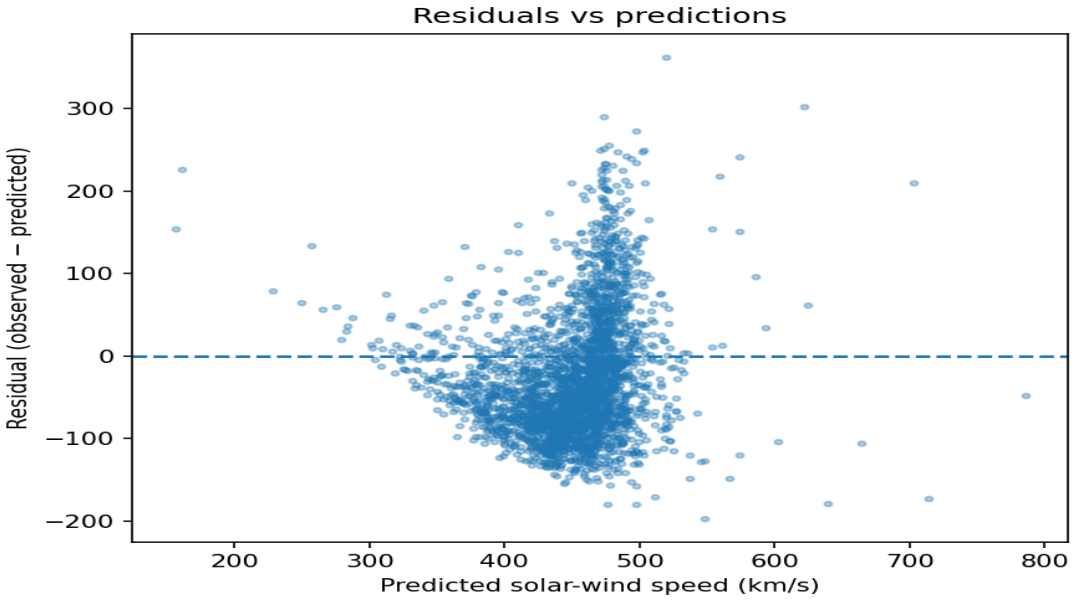
Diagnostics and interpretation Residual-versus-prediction and residual-distribution plots are included. The prediction-bin table summarizes mean residual, MAE, and residual spread across prediction ranges. Review these slices for systematic under/over-prediction and changing error variance.

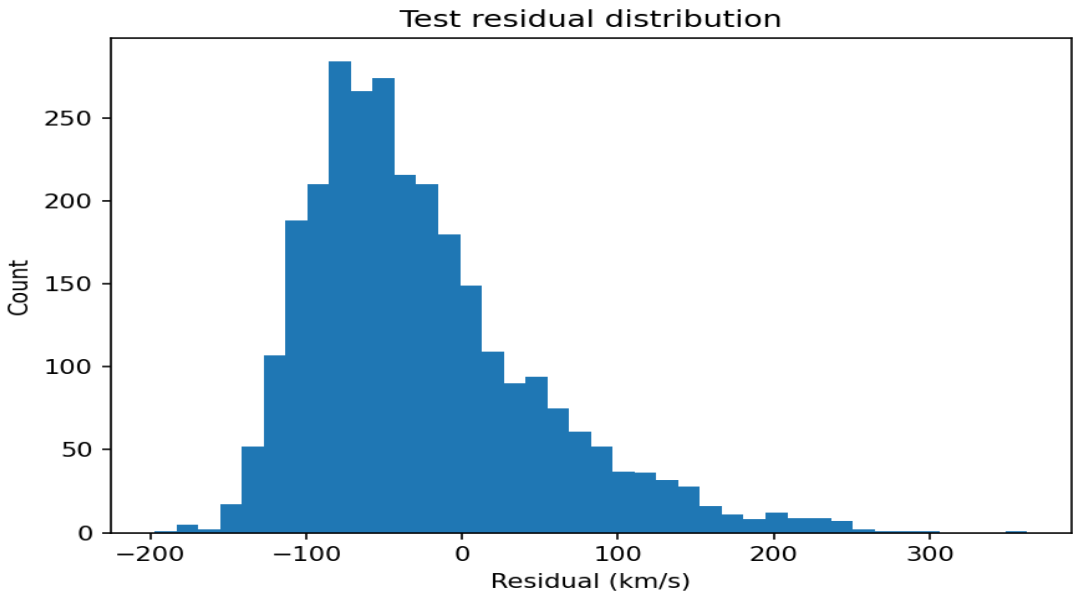
Claim boundary This is an interpretable predictive baseline, not a causal study. Linear associations may reflect shared solar-wind conditions, temporal dependence, measurement quality, or unobserved variables. The chronological test set is a stronger check than a random split but does not guarantee performance under future distribution shifts. Sentinel rules and positional column mapping should be independently checked against the

dataset documentation before extending the claims.

Reproduction Install packages in `requirements.txt`, place the frozen file at `data/omni_01_av.txt` (or beside the notebook), and run the notebook top-to-bottom. Scripts and raw outputs should be committed together with a dataset hash and the exact repository commit.






Test residual distribution
Count
0
50
100
150
200
250
−200
−100
0
100
200
300
Residual (km/s)